UNIT – III : Numerical Control & CNC Part Programming

1. Numerical Control (NC)

- **NC** = a form of programmable automation in which the operations of a machine tool are controlled by **numbers, letters and symbols** (coded instructions) that are fed to the machine. (EIA definition)
- Invented at MIT (1952, J. Parsons) for milling aircraft parts; now every machine tool is NC / CNC.
- The machine follows the program automatically: tool position, speed, feed, coolant, tool change.

Why NC?

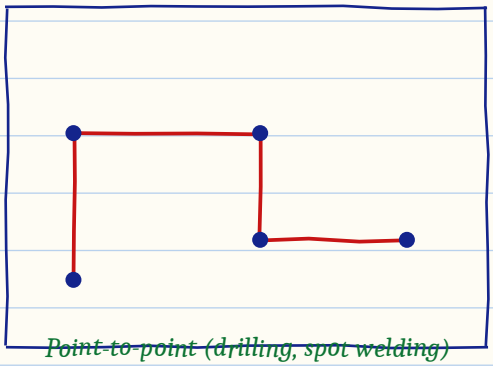
- High accuracy and repeatability, complex shapes, less setup time, small-batch (job-shop) flexibility, lower scrap, reduced operator skill & fatigue, less jigs and fixtures, easy design changes by editing the program.

Limitations

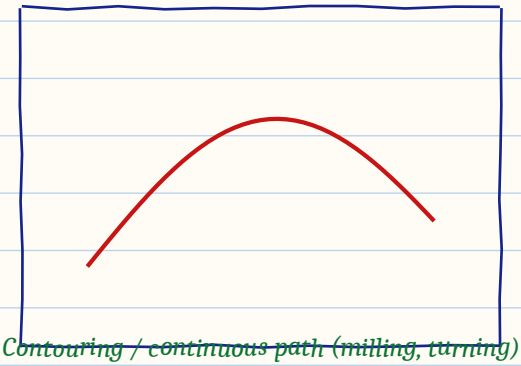
- High initial cost, skilled programmers and maintenance, not economical for very large batches (special-purpose machines are better), computer / electronic problems.

2. NC Modes (types of control / motion control)

- **Point-to-point (PTP) / positioning control**: tool moves to a position at rapid speed; machining is done only after reaching it (drilling, boring, tapping, spot welding, punching). Path between points is unimportant.
- **Straight-cut (line) control**: machining along a straight line parallel to an axis at controlled feed (simple milling, turning, facing).
- **Contouring (continuous path) control**: two or more axes move simultaneously with a coordinated speed ratio to cut any path – angled lines, arcs, curves (milling, lathe, profile cutting). Needs an **interpolator** (linear, circular, parabolic, cubic).



Point-to-point (drilling, spot welding)

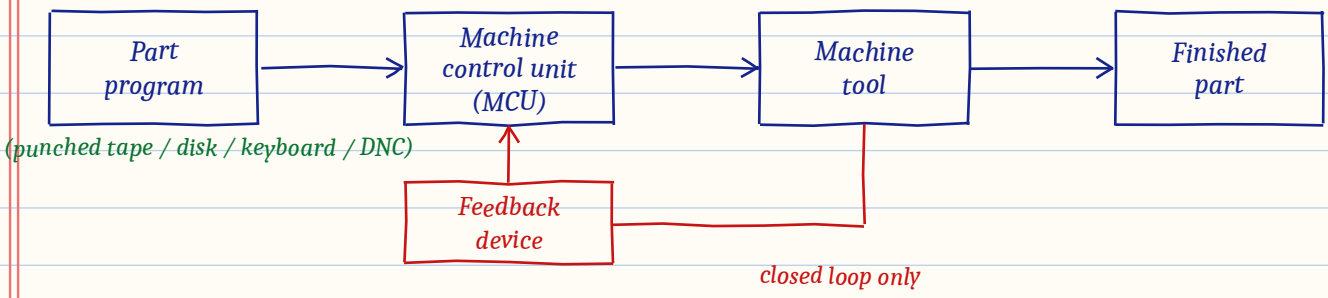


Contouring / continuous path (milling, turning)

- Programming modes: **absolute** (all dimensions measured from a fixed zero: G90) and **incremental** (each move measured from the previous point: G91).

3. Elements of an NC System

- Three basic elements:
  - **Program of instructions (part program)** – step-by-step commands in coded form (tape, disk, memory).
  - **Machine control unit (MCU)** – electronics that reads the program, decodes it and sends signals to drive the machine (data processing unit + control loops unit).
  - **Machine tool (processing equipment)** – the machine that does the cutting (with slides, spindle, table, drives, feedback devices).



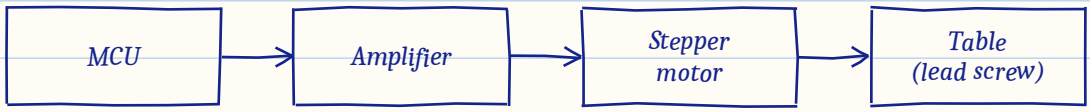
- **Data processing unit (DPU)**: tape reader, data buffer, decoding, interpolator, sends signals to the control loops unit.
- **Control loops unit (CLU)**: servo-drives for each axis, receives feedback, signals for spindle speed, coolant, tool change.

Open-loop and closed-loop systems

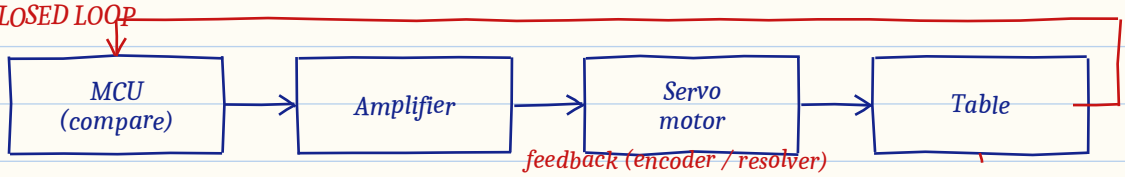
- **Open loop**: uses **stepper motor**, no feedback of actual position; simple, cheap, used for light loads; accuracy depends on the motor.
- **Closed loop**: **servo motor** (DC / AC) with **feedback** (encoder, resolver,

tachogenerator, LVDT); the actual position is compared with the commanded position and the error is corrected. Accurate but costly.

OPEN LOOP



CLOSED LOOP



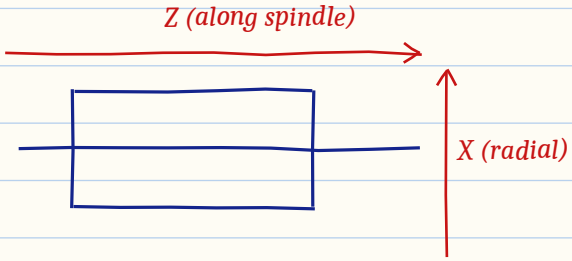
- Feedback devices: **encoder** (optical, incremental or absolute), **resolver** (analog, rotary transformer), **linear scale / inductosyn**, **tachogenerator** (speed).

4. NC Machine Tools and their Structure

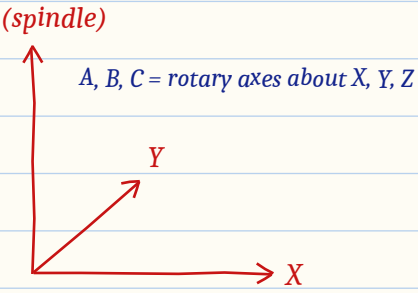
- Typical structure: **Bed / base** (cast iron or polymer concrete, rigid, damping) → **Column** → **Slides / guideways** → **Table** → **Spindle & headstock** → **Drives** → **Control panel** (MCU) → **Tool changer** and **coolant** system.
- Special design features of NC machines:
  - **High static and dynamic rigidity**, low thermal distortion (symmetrical design, cooling of spindle).
  - **Low-friction slideways**: hydrostatic ways or **linear motion (LM) guideways** with rolling elements; PTFE-coated (turcite) slideways to avoid stick-slip.
  - **Recirculating ball screw and nut** for table drive (efficiency about 90%, zero backlash with preloaded double nut).
  - **Backlash elimination**, quick-response servo drives, wide spindle speed range (variable speed AC / DC drive).
  - **Automatic tool changer (ATC)**, tool magazine, chip conveyors, enclosed safe guarding, in-process probing.

Axes of motion (ISO / EIA convention)

- **Z-axis** = along the spindle axis (positive direction = away from workpiece); **X-axis** = main horizontal axis; **Y-axis** by right-hand Cartesian rule; **A, B, C** = rotations about X, Y, Z.



CNC lathe (2 axes: X, Z)



Vertical milling (X, Y, Z)

NC lathe (turning centre)

- Axes X (cross slide) and Z (carriage); turret with 8 to 12 tools; chuck, tailstock; live tooling and C-axis in turn-mill centres.

NC milling machine

- Vertical or horizontal spindle; table moves in X, Y, knee / head in Z; 3-axis, 4-axis (rotary table) and 5-axis machines.

5. Machining Centre – types and features

- **Machining centre (MC)** = NC machine that can do **several machining operations** (milling, drilling, boring, tapping) on a workpiece in **one setup**, with **automatic tool changing** from a tool magazine and often an automatic pallet changer.

Types

- **Vertical machining centre (VMC)** – vertical spindle; for flat parts, dies and moulds; 3 axes.
- **Horizontal machining centre (HMC)** – horizontal spindle; for box-type parts; chips fall away; pallet changer, rotary table (B-axis).
- **Universal machining centre (UMC)** – 5-axis; tilting spindle or table; complex parts in one setup.
- **Turning centre / turn-mill centre** – lathe with live tools; **Gantry-type** machining centres for very large parts.

Features

- Automatic tool changer (ATC): 20 to 100+ tools, tool change in 2 to 8 seconds; tool identification by pocket number (random access) or tool code.
- **Automatic pallet changer (APC)** – load / unload while machining another pallet (reduces idle time).

- Rotary table indexing, probing for work-offset and tool-length measurement, high spindle speed (6000 to 30000 rpm), through-spindle coolant, chip conveyor.
- High productivity, accuracy, flexibility; basis of **FMS cells**.

6. Controls in NC / CNC

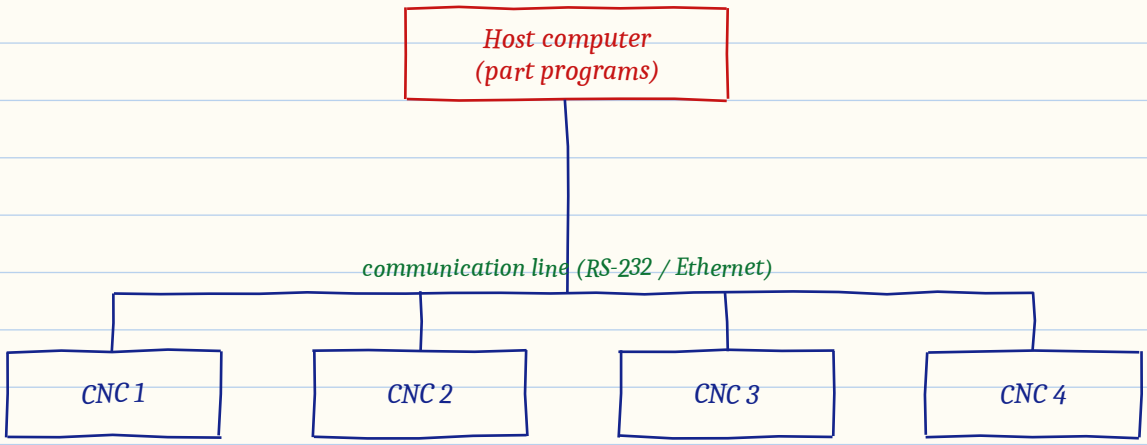
- Control types by hardware: **hard-wired NC** (fixed logic, early) → **soft-wired CNC** (microcomputer inside MCU; program kept in memory; can be edited at the machine).

CNC – Computer Numerical Control

- **CNC** = an NC system in which a dedicated **microcomputer** performs the basic NC functions by stored software (**executive program**).
- Features: program stored in RAM / memory, **editing at the machine**, graphic display and simulation, subprograms and macros (parametric programming), canned cycles, cutter-radius and tool-length compensation, diagnostics, communication ports, adaptive control possibility.
- Advantages over NC: more flexible, less hardware, easy upgrade by software, higher reliability, tape reader not required, program storage and reuse, accuracy through compensation of errors.

7. DNC – Direct (Distributed) Numerical Control

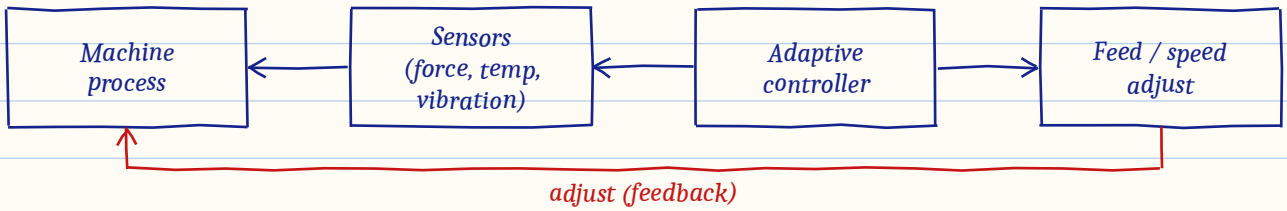
- **DNC** = use of a **central (host) computer** to control a number of NC / CNC machines by direct connection; the part programs are stored in the host and sent to the machines on demand (block by block or whole program).
- Two types:
  - **Direct DNC** – host computer is directly connected and controls the machines (no tape reader); if the computer fails all machines stop.
  - **Distributed DNC** – host sends programs to individual CNC machines with their own memory through a **network (LAN)**; machines continue even if the host is down.



- Components: central computer (programs), bulk memory, communication lines, machine tools (with MCU), software for program storage, download and tracking of production data.
- Advantages: **elimination of tape**, large program storage, easy program editing & transfer, production monitoring & data collection, scheduling and basis for FMS and CIM.

8. Adaptive Control Machining Systems

- **Adaptive control (AC)**: the machining conditions (feed, speed) are **automatically adjusted in real time** to the actual conditions of the process measured by sensors – force, torque, temperature, vibration, tool wear, power.
- Reason: conventional NC uses fixed feed / speed set conservatively; actual cutting varies due to hardness variation, tool wear, depth of cut, etc.



Types of adaptive control

- **ACC – Adaptive Control with Constraints**: a measured variable (cutting force, torque or power) is kept **below a limit** by adjusting the feed (or speed). Protects tool and machine; most common in practice.
- **ACO – Adaptive Control with Optimisation**: an **index of performance** (e.g. minimum cost per part or maximum production rate) is calculated on line and the feed /

speed are adjusted to optimise it (needs tool-wear / cost models – complex).

- **GAC – Geometric Adaptive Control**: compensates for geometric deviations (tool deflection, thermal growth) to maintain **dimensional accuracy and surface finish**.

Benefits

- Higher metal-removal rate, longer tool life, protects the machine & tool from overload, less operator involvement, better surface finish. Limits: costly sensors, sensor reliability in a harsh environment.

CNC Part Programming

9. Fundamentals of Part Programming

- **Part program** = set of instructions in a coded form that tells the machine what to do to make the part.
- Steps: study the drawing → select process, machine, tools, cutting data → fix the **work datum (zero point)** and clamping → decide tool path & sequence → write the program → enter on machine → **dry run / simulation** → prove out → production.
- Methods: **manual part programming** (G & M codes) and **computer-assisted part programming** (APT, Mastercam; geometry is defined, then the computer calculates the tool path and a post-processor converts it into machine code).
- Coordinate system: Cartesian right-hand rule. **Machine zero** (fixed by the maker) and **work zero / program zero** (set by the user, G54 to G59). **Reference point** is used for homing.
- Data to be decided: spindle speed $N = 1000\,V / (\pi D)$ rpm, feed (mm/min = $f \times N$), depth of cut, tool numbers & offsets, coolant.

10. NC Word and Block (Program Format)

- **Character** (letter, digit, symbol) → **word** (letter + number) → **block** (line = one operation) → **program**.
- **NC word** = address letter + numerical value. It gives one type of instruction.

Address	Meaning	Example
N	sequence (block) number	N010
G	preparatory function – mode of motion	G01 linear interpolation
X Y Z	coordinate positions	X50.0 Y25.0
I J K	arc centre (incremental) / arc parameters	I10 J0
F	feed rate (mm/min or mm/rev)	F120
S	spindle speed (rpm)	S1200
T	tool number	T03
M	miscellaneous (auxiliary) function	M03 spindle on
D / H	tool radius / length offset number	D01 H01

- **Block format**: *word address format* is standard (words may be omitted if

unchanged – modal). Example block:

N040 G01 X60.0 Y30.0 F150 S800 M03 ;

- **Modal** codes remain active until changed; **non-modal** act only in the block where they are written.
- Program starts with % / O-number, ends with M30 (end & rewind) or M02.

Important G codes (preparatory functions)

Code	Function	Code	Function
G00	rapid positioning	G01	linear interpolation (feed)
G02	circular interpolation clockwise	G03	circular interpolation counter-clockwise
G04	dwell (pause)	G17/18/19	XY / ZX / YZ plane selection
G20 / G21	inch / metric input	G28	return to reference point
G40	cutter compensation cancel	G41 / G42	cutter comp. left / right
G43 / G49	tool length comp. on / cancel	G54–G59	work coordinate systems
G80	cancel canned cycle	G81–G89	canned drilling cycles
G90 / G91	absolute / incremental	G94 / G95	feed per min / per rev

Important M codes (miscellaneous)

Code	Function	Code	Function
M00	program stop	M01	optional stop
M02	end of program	M03 / M04	spindle clockwise / counter-clockwise
M05	spindle stop	M06	tool change
M08 / M09	coolant on / off	M30	end of program & rewind

11. NC Nodes (Tool positioning, interpolation)

- **Nodes** = the points (end points of tool-path segments) that define the tool path; the controller joins them by interpolation.
- **Linear interpolation (G01)** – straight line between two nodes. **Circular interpolation (G02/G03)** – arc given by end point + centre (I, J) or radius R.
- Direction of arc: clockwise (G02) when looking from the +ve side of the axis perpendicular to the plane; **I, J** are the incremental distances from the start point to the arc centre in X and Y.

12. Canned Cycles

- **Canned cycle** = a single G code (with parameters) that performs a **frequently repeated multi-step operation** such as drilling, boring, tapping, so that program length is reduced.
- Typical drilling cycle sequence: rapid to R-plane → feed to depth Z → (dwell / peck) → rapid retract.
- Format: G8x X_ Y_ Z_ R_ Q_ P_ F_ ; (Z = depth, R = rapid-return (reference) plane, Q = peck depth, P = dwell time).

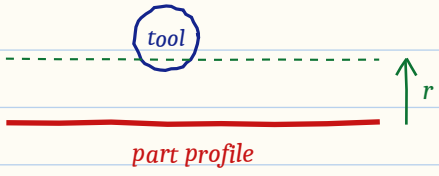
Code	Cycle	Use
G81	simple drilling	through holes, centre drill
G82	drilling with dwell	spot facing, counter-bore
G83	peck drilling (deep hole)	deep holes, chip breaking
G84	tapping	threaded holes
G85 / G86	boring (feed in – feed out / spindle stop)	finish bores
G73	high-speed peck	quick chip break

- Cancel the cycle with G80; the X, Y positions of the following blocks repeat the cycle at every position.
- On a **lathe**: G90 (turning cycle), G92 / G76 (threading cycle), G71 / G72 (rough turning / facing), G70 (finishing), G74 / G75 (grooving / peck drilling).

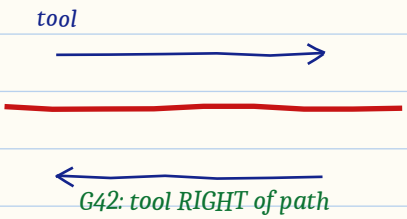
13. Cutter Radius Compensation (CRC / Cutter Diameter Compensation)

- The tool path programmed is that of the **part profile**; the control shifts the tool centre by the **tool radius** so that the edge cuts the programmed profile. Without CRC the programmer must calculate an offset path for every tool.

programmed path (tool centre) offset = r



G41: tool LEFT of programmed path



- **G41** = cutter compensation **left** (tool is to the left of the work profile in the direction

of travel – climb milling for a clockwise spindle).

- **G42** = cutter compensation **right**.
- **G40** = cancel compensation. The offset value is stored in the offset table (D01 etc.), so the same program works for a worn or re-ground cutter by just changing the offset.
- Steps: (1) approach with G00 away from the part, (2) turn on G41 / G42 with D-word in a **lead-in move** (a straight move at least equal to the radius), (3) machine the profile, (4) cancel with G40 in a lead-out move.
- Benefits: allows use of different cutter sizes, roughing and finishing with the same program, tolerance adjustment through offset change.

14. Tool Length Compensation

- Each tool has a different length; the control must know the **tool-length offset** from the spindle gauge line to the tool tip so the Z zero is correct for every tool.
- **G43** = tool length compensation **positive** (add H offset), **G44** = negative, **G49** = cancel. The H-number selects the stored value (H01 for T01).

```
G43  H01  Z100.0 ;      (tool 1 length is added to Z)
```

- Offsets are measured on the machine (touch-off or tool presetter) – the program need not change when the tool length changes. On lathes the tool offsets (X wear, Z wear) are called with the T word (T0101).

15. Manual Programs – Simple Components

(a) Milling – profile of a 60 × 40 rectangular block with a 10 mm hole at the centre (depth 10 mm)

- Work zero at lower-left corner top surface; tool: 10 mm end mill T01 for contour; 8 mm drill T02. Spindle 1000 rpm, feed 120 mm/min.

```
O1001 ;
N10 G21 G90 G17 G54 ; (metric, absolute, XY plane, work zero)
N20 T01 M06 ; (load end mill)
N30 S1000 M03 ; (spindle on CW)
N40 G00 X-10 Y-10 Z5 ; (rapid to start, above the work)
N50 G01 Z-10 F60 ; (plunge to depth)
N60 G41 D01 G01 X0 Y0 F120 ; (cutter comp left, lead-in)
```

N70 X60 ; N80 Y40 ; N90 X0 ; N100 Y0 ; (profile)

N110 G40 G00 X-10 Y-10 ; (cancel comp)

N120 Z50 M05 ; (retract, stop spindle)

N130 T02 M06 ; N140 S1200 M03 ;

N150 G81 X30 Y20 Z-12 R2 F80 ; (drill hole at centre)

N160 G80 G00 Z50 M05 ; N170 M30 ;

(b) Turning – step shaft: 40 mm bar, turn 30 mm dia for 40 mm length, then 20 mm dia for 20 mm (tool tip at X42 Z2)

O2001 ;

N10 G21 G90 G99 ; N20 T0101 ; N30 G96 S120 M03 ; (constant surface speed)

N40 G00 X42 Z2 ; N50 G01 X30 Z0 F0.15 ; N60 Z-40 ;

N70 X20 ; N80 Z-60 ; N90 X42 ; N100 G00 X100 Z100 ; N110 M30 ;

(c) Circular interpolation

- Quarter arc from (20, 0) to (0, 20) with centre at origin, counter-clockwise:

G03 X0 Y20 I-20 J0 F100 ; (or G03 X0 Y20 R20)

(d) Drilling a pattern of 4 holes using a canned cycle

G81 X20 Y20 Z-15 R2 F90 ; X60 ; Y60 ; X20 ; G80 ; (the cycle repeats at each position)

Computer Assisted Part Programming – APT

16. Computer-Assisted Part Programming

- In **computer-assisted programming** the programmer writes the program in a **high-level, English-like language**; the computer calculates cutter-location (CL) data and a post-processor makes the machine code. Used for complex 3D / 5-axis parts. Languages: **APT** (Automatically Programmed Tool, MIT 1956), ADAPT, EXAPT, COMPACT II, Pro-NC, and modern CAM (Mastercam, NX CAM).
- Four stages of APT processing: **(1) Input translation** (source program → machine-readable), **(2) Arithmetic (definition and motion) calculations**, **(3) Editing / CL file output (CLDATA)**, **(4) Post-processing** → NC program for a particular machine and control.
- The **part programmer** writes the part program; the **processor** (general, machine-independent) generates CLDATA; the **post-processor** (machine specific)

outputs G-code.

Source program → APT processor → CL file → Post-processor → NC part program → Machine.

17. APT Statements – the five types

(a) Geometry statements – define the elements of the part

- Format: symbol = geometry type / defining parameters.

P1 = POINT / 20, 30, 0 (point by coordinates)

L1 = LINE / P1, P2 (line through two points)

L2 = LINE / P1, PARLEL, L1 (parallel line) L3 = LINE / P1, PERPTO, L1

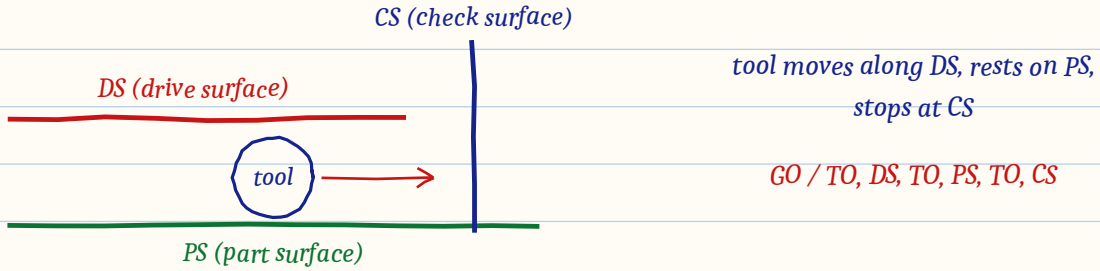
C1 = CIRCLE / CENTER, P1, RADIUS, 15 (circle)

PL1 = PLANE / P1, P2, P3 (plane through three points)

- Many forms are available: line by slope (LINE / P1, ATANGL, 45), circle tangent to lines (CIRCLE / YSMALL, L1, L2, RADIUS, 10), ellipse, cone, cylinder, sphere, vectors, matrices, patterns (PATERN).

(b) Motion statements – tool movement

- **Point-to-point**: FROM / P0 (start point); GOTO / P1 ; **GODLTA / dx, dy, dz** (incremental move).
- **Continuous path**: uses three surfaces – **drive surface (DS)**, **part surface (PS)** and **check surface (CS)**. The tool is kept tangent to the DS and PS and stops at the CS.
- Start-up: **GO / TO, DS, TO, PS, TO, CS** ; then moves: **GOLFT / L1, TANTO, C1** (go left along L1 tangent to C1), **GORGT, GOFWD, GOBACK, GOUP, GODOWN**.
- Tool direction terms: TO (tool touches surface), ON (tool centre on surface), PAST (tool beyond the surface), TANTO (tangent to).



(c) Post-processor statements – machine & control data

- **MACHIN / name, number** (select the post-processor: e.g. MACHIN / FANUC, 3).
- **COOLNT / ON, OFF, MIST, FLOOD**, **SPINDL / 1000, CLW, (CCLW)**, **FEDRAT / 120**, **RAPID**, **LOADTL / 1** (tool change), **TURRET**, **END**, **FINI** (end of the program), **CUTTER / 10** (tool diameter), **TOLER**, **INTOL / OUTTOL** (tolerances), **OFSTNO**.

(d) Auxiliary statements – supply non-geometric information

- **PARTNO** (title / name), **REMARK**, **INSERT**, **PRINT**, **CLPRNT**, **PPRINT**, **INTOL**, **OUTTOL**, **COOLNT**, **TLON**, **STOP**.
- Their position: PARTNO is the first statement; FINI is the last.

(e) Macro statements – subprograms (repeated shapes)

- **Macro** = a named group of APT statements with parameters that is **defined once and called many times** – like a subroutine.

HOLE = MACRO / X, Y, DEPTH (macro header with arguments)

GOTO / X, Y, 5 ; GODLTA / 0, 0, -DEPTH ; RAPID ; GODLTA / 0, 0, DEPTH ;

TERMAC (end of the macro)

CALL / HOLE, X = 20, Y = 30, DEPTH = 12 (call with actual values)

18. APT Program for a Simple Component

Example – profile milling of a plate (rectangle 80 × 50) and a drilled hole, cutter dia 10 mm

PARTNO PLATE (title)

MACHIN / FANUC, 3 (post-processor)

CLPRNT ; UNITS / MM ; CUTTER / 10 (print CL data, metric, tool dia)

P0 = POINT / -20, -20, 5 (tool start point)

L1 = LINE / 0, 0, 0, 80, 0, 0 (bottom edge) L2 = LINE / 80, 0, 0, 80, 50, 0

L3 = LINE / 80, 50, 0, 0, 50, 0 (top edge) L4 = LINE / 0, 50, 0, 0, 0, 0

PS = PLANE / 0, 0, 1, 0 (part surface z = 0) P5 = POINT / 40, 25, 0

FROM / P0 ; FEDRAT / 120 ; SPINDL / 1000, CLW ; COOLNT / ON

GO / TO, L1, TO, PS, TO, L4 (start-up statement)

GORGT / L1, PAST, L2 ; GOLFT / L2, PAST, L3 ; GOLFT / L3, PAST, L4 ; GOLFT / L4, PAST, L1

GOTO / P0 ; RAPID ; GOTO / P5 ; GODLTA / 0, 0, -12 ; GODLTA / 0, 0, 12 ; SPINDL / OFF ;

COOLNT / OFF

FINI

- **Result**: processor creates CL data → post-processor converts into G / M codes for the

selected machine.

Advantages of APT / computer-assisted programming

- Easy for complex geometry (3D, 5-axis), less error, calculations done by computer, tool path plot / simulation, reusable macros, one program for many machines (different post-processors).

Limitations

- Needs a computer and a trained programmer, geometry has to be defined in the language, cost of software, no direct link to a CAD model (older systems) – solved in modern CAD/CAM.

19. Quick Revision

NC elements: part program + MCU + machine tool. NC types: PTP, straight-cut, contouring. Loops: open (stepper) / closed (servo + feedback).

- CNC = microcomputer inside MCU; DNC = host computer controls many machines; adaptive control = ACC, ACO, GAC.
- NC word = letter + number; block = line; modal vs non-modal; G = preparatory, M = miscellaneous.
- Canned cycles G81–G89 reduce program length; G41/G42/G40 cutter radius compensation; G43/G49 tool length compensation.
- APT: 5 statement types – geometry, motion, post-processor, auxiliary, macro. Motion uses DS, PS, CS.

Likely exam questions

- Define NC and explain the elements of an NC system with a block diagram.
- Differentiate between open-loop and closed-loop NC systems. Explain NC, CNC and DNC.
- Describe the features and types of machining centres.
- What is adaptive control? Explain its types with applications.
- Explain NC word, block format and the commonly used G and M codes.
- Write a manual part program (with canned cycle / compensation) for the given component.
- Explain geometry, motion, post-processor, auxiliary and macro statements of APT with examples.